

DANIEL OMONIGUHO OKENE

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RESEARCH INTEREST

Computational Thermo-Fluids, Multi-physics modelling, Fluid Mechanics, Turbulence Modelling, Aerodynamics, Computational Engineering, Energy and Propulsion, and Machine Learning Applications.

EDUCATION

University of Benin, Benin City, Edo State, Nigeria

01/2020 – 04/2025

Bachelor of Engineering (BEng), Mechanical Engineering

First Class Honors - CGPA: 4.69/5.0 (3.79/4.0; WES iGPA | 3.828/4.0; Scholaro)

Class Rank: 1 out of 109.

Thesis Title: Computational Fluid Dynamics (CFD) Analysis of Cavitation Effects on Centrifugal Pump Performance and Flow Patterns in Oil and Gas Applications Using ANSYS.

RESEARCH EXPERIENCE

Graduate Research Collaborator

07/2025 – Present

Next Energy Systems Technologies (NEXT) Research Hub

Supervisor: Prof. Ighodaro

- Conducting off-design modelling and thermodynamic performance assessment of Combined Cycle Power Plants, analyzing energy and exergy efficiencies across major components under baseload and part-load conditions.

Undergraduate Research Assistant, University of Benin

05/2024 – 04/2025

Thermal-Fluid Lab

Supervisor: Dr. Ikechukwu Owunna

- Led the team that performed CFD investigation on crude-oil cavitation effects in centrifugal pumps using ANSYS Fluent, applying the VOF multiphase model, RNG k- ϵ turbulence model, and Schnerr-Sauer bubble model.
- Conducted a series of simulations analyzing the influence of operating parameters, including inlet pressure, wall roughness, and backflow intensity. Identified that backflow and roughness values greatly intensified cavitation behavior and performance degradation.

Undergraduate Research Assistant, University of Benin

09/2024 – 03/2025

Materials and Ballistic Laboratory

Principal Investigator: Prof. Ighodaro

- Created molds for making kenaf-fiber reinforced composite, carried out tensile, flexural, and impact tests on the composite.
- Developed and validated the composite properties and models using the two-step homogenization techniques and experimental test data to capture composite behavior for impact simulation.
- Simulated high-velocity ballistic impacts on kenaf fiber-reinforced composites using ANSYS AUTODYN, focusing on stress distribution, deformation, and energy absorption.

Summer Research Intern

06/2024 – 08/2024

The European Organization for Nuclear Research [CERN] – Geneva, Switzerland

- Collaborated with engineers and professors from the University of Federico II, Italy, on the structural dynamics analysis of the decay vessel for the proposed Search for Hidden Particles (SHiP) experiment.
- Modelled the latest Decay Vessel configuration using ANSYS SpaceClaim. Replicated a wave-guided modal analysis using ANSYS Mechanical, referencing MATLAB code from a research paper. Computed mode shapes, frequencies, and plotted dispersion curves based on analysis results. Applied the same methodology to the Decay Vessel geometry. Findings were uploaded to CERN's server. [\[Internship Report\]](#)

PUBLICATION

- Owunna, I., **Okene, D.**, Madu, C., Nwabueze, B. (2025). A Numerical Approach to Studying Crude Oil Cavitation in Centrifugal Pumps. (pgs. 144 – 166, **Book of Proceedings, NIPES 3rd International Conference**).
- Ighodaro O., Oluwole, A., **Okene, D.**, Madu, C., (2025). Off Design Modelling and Performance Assessment of The Afam VI Integrated Combined Cycle Power Plant. *Applied Energy (Elsevier)*, **In Review**.

- Ighodaro, O., Egware, H., Nwankwo, E., Orumwense, E., Kwasi-Effah, C., Abdulazeez, M., Amune, J., Okpako, O., Esimike, T., **Okene, D.**, Madu, C., Dan-Odhomo, J., Ogunmola, B. (2025). Multiscale Homogenisation and Mechanical Characterisation of Kenaf 2D Woven Fabric Composites with Ballistic Relevance. *Progress in Engineering Science* (Elsevier), In Review.

CONFERENCES AND POSTER PRESENTATION

- Presented my research work titled: “Investigation on the Structural Dynamics of the Decay Vessel for the SHiP Experiment” at the CERN summer student poster session. 26th July 2024. [[Poster](#)]
- “A Numerical Approach to Studying Cavitation in the Oil and Gas Industry” Presented at the National Institute of Professional Engineers and Scientists 3rd International Conference, “Converging of Emerging Technologies: Redefining the boundaries of science and technology”, included in the book of abstracts. University of Benin, 9-11 April 2025.

PROFESSIONAL EXPERIENCE

FEA Analyst and Developer, Africa Makes - Lagos 08/2024 – Present

- Teaching the theoretical foundations of FEA to software developers, ensuring accurate implementation of numerical methods in solver development. Contributing to the development of Africa’s first cloud-based FEA solver (*Enitan*) using Python, validating numerical outputs, and ensuring modular and extensible code architecture.
- Serving as a QA tester for the Enitan Studio and Classroom applications, conducting functionality testing, and providing technical feedback to enhance user experience and solver accuracy.

Facilities Engineer Intern, Chevron Nigeria Limited – Lagos 01/2024 – 05/2024

- Monitored the reconstruction of tank farms in OKAN under the supervision of a project engineer. Learned and applied knowledge of P&ID and process engineering to support project activities.
- Participated in Process Hazard Analysis (PHA) and Hazard and Operability (HAZOP) sessions to ensure operational safety. Attended budget defense meetings with the National Upstream Investment Management Services (NUIMS).

TEACHING EXPERIENCE

Graduate Assistant, Department of Mechanical Engineering, University of Benin. 05/2025 – Present

- Assisting in teaching, preparing course material, grading tests, and exams for Fluid Mechanics III – V (MEE461, MEE561, and MEE562). Additionally, I anchor tutorial sessions and advise final-year undergraduates on their projects.

Academic Tutor, University of Benin 02/2023 – 04/2025

- Organized weekly tutorials on Engineering Mechanics, Thermodynamics, and Fluid Mechanics to over 100 undergraduate students across several departments, contributing to a 70% improvement in test and exam performance.

Tutor, KTTC Tutorials - Lagos. 06/2019 – 03/2020

- Taught Physics, Mathematics, and Chemistry to over 150 WAEC and JAMB candidates, achieving a record-breaking 90% pass rate in the tutorial center.

COMPETITION AND PROJECT

NADDC, Maiden Campus Shuttle Electric Vehicle Design Competition 04/2025 – 07/2025

- Member of the Vehicle Body and Structural Design Team, where I carried out research on EV skateboard chassis and performed aerodynamic CFD simulations to determine the best vehicle body configuration from our box fish model.

Autodesk Designathon 2024 06/2024

- Member of Team Chevy, where I modelled and designed a waste collector vehicle that operates both on land and water using Fusion 360. [[Certificate of Participation](#)]

VOLUNTEERING AND LEADERSHIP EXPERIENCE

Volunteer, Bedrock Human Development Foundation 08/2022 – Present

- Coordinated the logistics team for the 9th UNIBEN Marathon Competitions and the second edition of the academic scholarship examination, ensuring smooth participant movement and materials distribution.

Volunteer, Design Lead, NIPES 3rd International Conference 02/2025 – 04/2025

- Designed the conference banner, backdrop, jotters, and booklets; and ensured high-fidelity outputs ahead of schedule.

Director of Academics, Nigerian Institute of Mechanical Engineers, UNIBEN Chapter. **03/2023 – 09/2023**

- Collaborated with over 10 scholars across different levels to organize tutorials for 150+ students, significantly boosting departmental academic performance. Initiated and coordinated scholarship and career development webinars, enhancing students' access to opportunities [Students won over **\$6000** in scholarships].

Volunteer, Graphics Designer, Nigerian Institute of Mechanical Engineers, UNIBEN Chapter **09/2021 - 02/2023**

- Designed flyers, posters, banners, and print media for webinars and academic programs, maintaining a consistent brand identity for the student chapter.

Team Lead and Brand Designer, AIESEC in Benin **03/2021 – 04/2022**

- Led a team of 15 designers in curating templates and assets for campaigns and events, strengthening the Local Committee's visual identity. Directed designs for Finance Week and YALI Collaboration Events, contributing to broader regional visibility and partner engagement.

Academic Director, Christ Apostolic Church Youth Fellowship (UNIBEN/UBTH) **08/2023 – 05/2024**

- Organized academic tutorials for over 100 fellowship members, leading to measurable improvements in academic performance. Designed and implemented academic mentorship strategies and provided counseling to support student growth and resilience. Additionally, I conducted the inaugural interdepartmental quiz competition.

HONOURS AND AWARDS

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| • Best Graduating Student, Department of Mechanical Engineering, University of Benin | 04/2025 |
| • Best Undergraduate Final-Year Research Project, Department of Mechanical Engineering, UNIBEN | 04/2025 |
| • MTN Science and Technology Scholarship Recipient | 09/2023 |
| • The Youth Center Academic Scholarship Recipient | 06/2023 |
| • Midwestern Oil and Gas Company Scholarship Recipient | 02/2022 |
| • Bedrock Foundation Scholarship Recipient | 07/2022 |
| • 2X John Asiruwa Excellence Award for Best 200 & 300 Level Mechanical Engineering Student | 05/2023 |
| • Award of Selfless Leadership – Christ Apostolic Church Youth Fellowship UNIBEN/UBTH | 02/2025 |
| • Award of Recognition – Nigerian Institute of Mechanical Engineers. UNIBEN Student Chapter | 10/2020 |

TRAININGS AND CERTIFICATIONS

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| • Turbulence Modelling <i>IIT Madras</i> | 07/2025 - 10/2025 |
| • Computational Fluid Dynamics Using Finite Volume Method <i>IIT Madras</i> | 07/2025 - 10/2025 |
| • Introduction to High Performance and Parallel Computing <i>University of Colorado Boulder</i> | 05/2025 - 07/2025 |
| • Machine Learning Specialization <i>Stanford University</i> | 10/2024 - 04/2025 |
| • Aspire Leaders Program <i>Aspire Institute and Harvard Business School</i> | 08/2025 - 10/2025 |
| • Fundamentals of Manuscripts Preparation <i>Elsevier Research Academy</i> | 09/2025 |
| • SolidWorks CAD Design Associate (CSWA) <i>Dassault Systèmes</i> | 09/2024 |
| • Introduction to Ansys Fluent <i>Ansys</i> | 01/2024 - 02/2024 |
| • Computer Aided Engineering <i>University of Abuja – HEPSSA –Royal Academy of Engineering</i> | 03/2023 - 03/2024 |

PROFESSIONAL SKILLS

- Technical: AutoCAD, Fusion 360, SOLIDWORKS, ANSYS, HyperWorks Suites, MATLAB, C, C++, Python, Microsoft Suites, Laboratory Techniques.
- Soft Skills: Critical Thinking, Research, Data Analysis, Creativity, Problem Solving, Collaboration, Communication, and Time Management.

PROFESSIONAL MEMBERSHIPS

- CERN Alumni Network.
- Nigeria Institute of Mechanical Engineers, NiMechE.
- National Institute of Professional Engineers and Scientists, NIPES.